Waiting for the Stars: Optimal Routes Through the Real, Moving Solar Neighbourhood

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Abstract

How far apart do stepping stones have to be before a civilisation can hop across the Galaxy, and does the motion of the stars change that answer? We treat the 40,229 stars of the Gaia Catalogue of Nearby Stars within 50 pc as a graph with full 6D phase-space information and solve the routing problems exactly. With stars frozen in place, reach percolates sharply. A ship whose single hop is limited to r reaches almost nothing below $r \approx 2.12$ pc and half of all stars within 25 pc at $r_{50} = 2.16$ pc (95% CI 2.15–2.17). That is about twice the median nearest-neighbour spacing and 5% above a Poisson field of the same density. With Gaia alone the Sun is trapped in a 3-star pocket whose way out is a single 2.27 pc hop. Two nearby brown dwarfs that Gaia cannot see open it. Once the stars move and colonies may wait for a neighbour to drift within range, the problem becomes an exact earliest-arrival search on a temporal graph whose edges are open on a single closed-form time window. Waiting cuts the range needed to reach half the stars by 20% within 100 kyr and by 52% within 500 kyr. Below the static threshold, expansion is carried by the stars themselves at about 21 km/s. We map propulsion to range: a fusion ship at 3.8% of c must survive 184 yr per hop to percolate, or 88 yr if it can wait. Refuelling only at known exoplanet hosts needs a 3.4× longer tank. Photon sails that brake on starlight reach far fewer stars, because most targets are too dim to stop them. Optimised multi-probe tours are 7–9% shorter than the nearest-neighbour rule of earlier models.

1 Introduction

Interstellar distances are measured in parsecs: one parsec is 3.1×10^{13} km, and even the nearest star, Proxima Centauri, is 1.30 pc away. Any civilisation that spreads beyond its own system therefore moves *star to star*, and the natural mathematical object is a graph. The stars are the nodes, a hop is an edge, and the engineering limits of a ship (fuel, speed, how long the hardware or crew survives) cap the length of the edges it can use.

This framing goes back to the Fermi-paradox literature [12, 22]. Those studies asked how long a fleet of probes would take to cover the Galaxy, on synthetic star fields with the stars held fixed [3, 7]. Two things have changed since then. First, Gaia now gives positions, proper motions and, for most nearby stars, radial velocities [11, 10], so the real 6D star field can be routed instead of a random one. Second, Carroll-Nellenback et al. [5] argued that stellar motions alone can carry settlement across the Galaxy. That claim was made with a statistical front model, not with the routes themselves.

We ask four concrete questions on the real catalogue.

1. **Static reach.** With stars frozen, what hop range r is needed to reach a given fraction of the neighbourhood from the Sun, and where are the bottlenecks?
2. **Moving stars.** If a colony may *wait* for a neighbour to drift within range, how much smaller can r be? This is an exact time-dependent shortest-path problem, and it is the main contribution.
3. **Engineering.** Which propulsion technologies, per-hop lifetimes, refuelling rules and pilot heuristics turn that range into a route?
4. **Exploration.** How much do optimised multi-probe tours gain over the greedy rule used by earlier models?

Contributions. (i) An exact reduction of minimum-range routing to the bottleneck of the Euclidean MST, applied to 40,229 Gaia stars, with astrometric bootstrap errors, a Poisson null model and a test for unseen brown dwarfs. (ii) A closed-form temporal graph for linearly moving stars, in which every edge is open on exactly one time window, and an exact earliest-arrival Dijkstra over it. Waiting for stars lowers the percolation range by up to 58% within 1 Myr. (iii) A propulsion-to-range map, an exact fuel-bottleneck algorithm for refuelling only at exoplanet hosts, and a luminosity-limited photon-sail network. (iv) A reproduction of the classic probe-exploration model on real stars. Every number in the text is generated from the code and checked by unit tests against brute-force references.

2 Background and related work

Galactic exploration models. Hart [12] and Newman & Sagan [22] estimated colonisation timescales analytically. Bjørk [3] simulated probes visiting about 40,000 stars with a nearest-neighbour rule. Cotta & Morales [7] cast each probe's itinerary as a travelling salesman problem and shortened tours by about 10%. Forgan et al. [9] and Nicholson & Forgan [23] added probe dynamics and gravitational slingshots. All of these used static, synthetic star fields.

Moving stars. Carroll-Nellenback et al. [5] showed that stellar motions let a settlement front spread even when ships have short range. They modelled the front statistically, not as explicit routes on measured stars. Gaia has made stellar encounters with the Sun measurable: Bailer-Jones et al. [2, 1] find that Gliese 710 will pass within about 0.05 pc (∼10,000 au) of the Sun in about 1.3 Myr. We use that encounter as an end-to-end check of our kinematics.

Trajectory competitions. GTOC X, "Settlers of the Galaxy" [24], asked teams to settle a synthetic galaxy of 100,000 moving stars with fuel-limited ships and rewarded large, even settlements. Winning approaches combined tree search with heuristics. It is the closest problem in spirit, but its stars were synthetic and it was scored for the competition rather than built to measure physical quantities.

Sails. Heller & Hippke [14] and Heller, Hippke & Kervella [15] showed that a light sail can shed speed using the target star's radiation and gravity, and computed routes through α Cen to Proxima. Breakthrough-Starshot-style laser launch [19] supplies the departure speed.

Algorithms. Minimax (bottleneck) paths lie on the minimum spanning tree. Continuum percolation of random spheres in 3D has a known critical density [18]. Earliest-arrival search on time-dependent networks is exact with Dijkstra when edges are FIFO [6, 16]. A* [13], greedy geographic routing and its local-minimum failures [4], and 2-opt [8] are standard tools.

Gap. To our knowledge no prior work routes the real Gaia 6D neighbourhood with exact time-dependent shortest paths, or connects that analysis to propulsion, refuelling and sail-braking constraints.

3 Data and model

Catalogue. We select every Gaia DR3 source with parallax $\varpi > 20$ mas (inside 50 pc) and $\varpi/\sigma_\varpi \geq 10$, and keep only members of the Gaia Catalogue of Nearby Stars (GCNS) [10]. This step matters. About 18% of the raw parallax cut is not in the GCNS, and those rows have a median tangential speed of 2.6 km/s, which is the signature of distant background sources with spurious parallaxes. With them included, star density rises with distance and stars appear to fall toward the Sun. After the cut the density is flat at 0.077 pc^{-3} in every 10-pc shell. Four bright stars that saturate Gaia (α Cen A/B, Sirius A, Procyon A) and the two nearest brown dwarfs (Luhman 16AB [20] and WISE 0855−0714 [21]) are added from the literature. The result is 40,229 objects. 73% have a radial velocity, from Gaia DR3 or the GCNS literature compilation. 602 are confirmed exoplanet hosts (NASA Exoplanet Archive, matched by Gaia DR3 identifier).

Missing radial velocities. We fit a Gaussian velocity ellipsoid $\mathcal{N}(\mu, \Sigma)$ to stars with RVs and draw each missing RV from that Gaussian *conditioned on the star's measured tangential velocity*. Drawing RVs unconditionally breaks the coherence of the velocity field and makes faint stars converge on the Sun. In our tests that error tripled the star count inside 15 pc within 1 Myr. A unit test checks that the conditional fill is unbiased on a synthetic field.

Kinematics. Positions (pc) and velocities (pc/Myr) follow from standard astrometric formulae and agree with `astropy` to 10^{-10}. Stars move on straight lines. Over $\lesssim 1$ Myr inside 50 pc, Galactic curvature bends paths by far less than the inter-star spacing, the same regime used for Gaia encounter studies. With linear motion, Gliese 710 passes the Sun at 0.051 pc (10,600 au) in 1.29 Myr, in line with the Gaia DR3 encounter analysis [1].

Ships. A ship refuels at every star it visits, unless stated otherwise, and has a maximum single-hop range r. Range is the natural single number because for a coasting ship it is the product of cruise speed and the time the craft must survive in transit (Section 6). We score stars within an evaluation radius well inside the catalogue: 25 pc for frozen stars and 15 pc for moving stars. The outer shell acts as a boundary buffer.

4 Frozen stars: a sharp percolation threshold

Exact reduction. With stars fixed, the smallest range that reaches star j from the Sun is the *bottleneck* of the best path: the minimum over paths of the largest hop. On a complete Euclidean graph this minimax path runs along the minimum spanning tree (MST). One MST, built from a k-NN graph and verified against the dense MST, gives the bottleneck of all 40,229 stars at once.

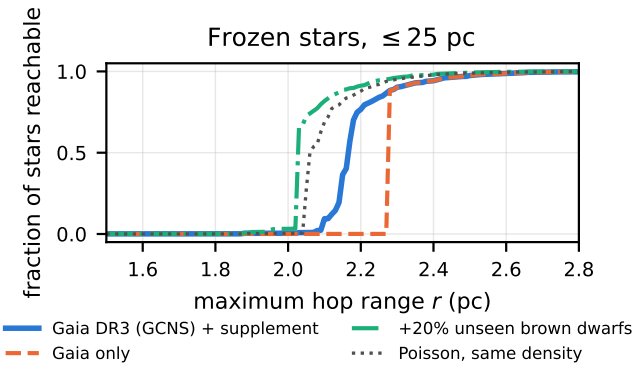


Figure 1: Fraction of the 5,117 stars within 25 pc reachable from the Sun with frozen stars, against the maximum single hop r. The transition is abrupt. With Gaia alone (dashed) the Sun's 3-star pocket holds everything until r reaches 2.27 pc. Unseen brown dwarfs (dash-dot, +20%) and a Poisson field of equal density (dotted) shift the threshold slightly lower.

The reachable fraction at range r is the fraction of stars whose bottleneck is $\leq r$.

The threshold. Reach jumps from almost nothing to almost everything over a narrow band of r (Figure 1). Ten percent of the stars within 25 pc are reachable at $r_{10} = 2.12$ pc, half at $r_{50} = 2.16$ pc (bootstrap 95% CI over Gaia astrometric errors: 2.15–2.17 pc), and 90% at $r_{90} = 2.30$ pc. The median nearest-neighbour distance is 1.18 pc, so a ship must out-range the typical spacing by a factor of about two. The threshold barely depends on the evaluation radius: r_{50} lies between 2.16 and 2.16 pc for radii of 10–25 pc. This is percolation. A Poisson field of the same density gives $r_{50} = 2.07$ pc, close to the infinite-volume continuum value of 2.04 pc [18]. The real neighbourhood is 5% harder to cross than random, a mild cost of its clustering.

The Sun's pocket. With Gaia alone, the bottleneck for almost every star is the same hop. The Sun, Proxima, Barnard's star and Ross 154 form an island whose shortest way out is a 2.27 pc jump from Ross 154. Adding Luhman 16 and WISE 0855, the two nearest brown dwarfs, both invisible to Gaia, opens the pocket, and the threshold becomes a property of the whole field. The lesson is that the *least visible* objects can be the most important stepping stones. Injecting a random population of unseen brown dwarfs at 20% (30%) of the stellar number, roughly the ratio in the 20-pc census [17], lowers r_{50} to 2.02 (1.97) pc, a 7% reduction at 20%.

Route shape. Just above threshold, at r_{90}, the shortest routes are long detours. The median route flies 2.5× the straight-line distance in 28 hops. Section 6 shows that this regime defeats simple greedy pilots.

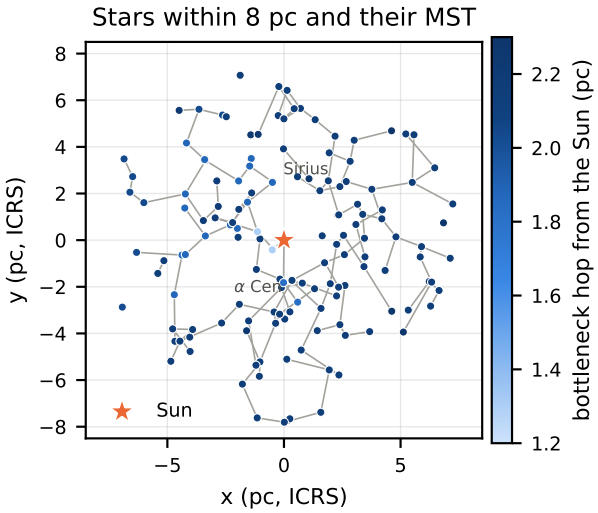


Figure 2: The 3D minimum spanning tree of stars within 8 pc, projected on the ICRS x–y plane, coloured by each star's bottleneck hop from the Sun. Nearly every star shares the same bottleneck.

5 Moving stars: waiting is a resource

A temporal graph with closed-form edges. Under linear motion the separation of stars i and j is $d_{ij}(t) = |p_{ij} + w_{ij}t|$, where p_{ij} is the relative position at $t = 0$ and w_{ij} the relative velocity. Since d_{ij}^2 is a convex quadratic in t, a hop of range r is open on *exactly one* interval $[a_{ij}, b_{ij}]$, found in closed form. A ship cruising at v intercepts a moving target after τ solving $|p + w\tau| = v\tau$. We open the edge when $d_{ij}(t) \leq r(1 - |w|/v)$, which guarantees a flown distance $\leq r$. A colony at i, ready at time t, leaves at $\max(t + s, a_{ij})$ if that is $\leq b_{ij}$, where s is an optional settling delay. Leaving later never arrives earlier, so the network is FIFO and Dijkstra over arrival times is exact [16]. Candidate pairs come from time-sliced KD-tree queries. A unit test checks that no pair whose separation ever drops below r is missed. A second test checks the arrival times against a Bellman–Ford fixed point. One run per r yields reach at every horizon T at once. With $r \leq 2.25$ pc and $T = 1$ Myr the graph has 817,591 candidate pairs.

Waiting lowers the threshold substantially. Figure 3 is the central result. If a colony may wait up to 100 kyr, the range needed to reach half the stars falls from 2.16 to 1.73 pc, a 20% reduction (bootstrap range 19–20%). The reduction is 35% within 200 kyr, 52% within 500 kyr ($r_{50} = 1.04$ pc) and 58% within 1 Myr. Our falsification criterion, a reduction below 5% at every horizon up to 10 Myr, is decisively rejected. Stellar motion is not a correction to interstellar routing. Over 10^5-year horizons it dominates.

The stars carry the front. For sub-threshold ranges the expansion speed is set by stellar motion, not by the

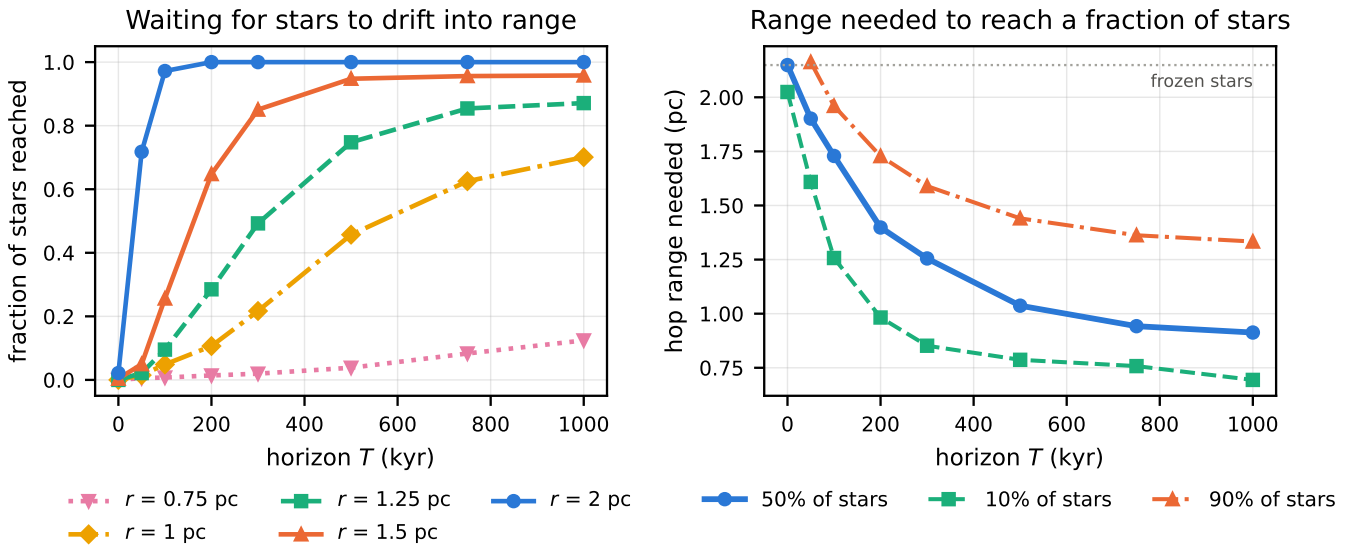


Figure 3: Moving stars, 1,070 stars within 15 pc, ships at 0.1 c. **Left:** fraction reached by horizon T for several hop ranges. Below the frozen threshold, reach grows steadily as stars drift into range. **Right:** range needed to reach 10/50/90% of stars by time T. The band is the spread over 3 astrometric bootstrap draws. The frozen-star value is the dotted line.

ship. With $r = 1$ pc, half the stars within 15 pc are reached at an effective speed of 22 pc/Myr (21 km/s), comparable to the local velocity dispersion. With $r = 1.5$ pc it is 76 pc/Myr. Naively extrapolated across a 30-kpc Galaxy, these give 1.4 and 0.4 Gyr, far shorter than the Galaxy's age. This is the Carroll-Nellenback et al. [5] conclusion, obtained here from explicit routes on measured stars.

What slows it down. *Settling time* matters most. At $r = 1.5$ pc and $T = 200$ kyr, 65% of stars are reached with instant relaunch, 44% if each colony needs 10 kyr before it can launch again, and 2% with 100 kyr. *Ship speed* matters less than one might think: 45% at 0.01 c versus 65% at 0.1 c. Only at 0.001 c (300 km/s, comparable to stellar motions) does reach collapse, to 3%.

Boundary check. Stars that would enter from beyond the catalogue are missing, so a catalogue-limited run *underestimates* the benefit of motion. The count of catalogued stars inside 15 pc stays at 104% of its initial value at 500 kyr and drops to 65% at 1 Myr. Results beyond 500 kyr are therefore conservative. The full configuration uses the 100-pc catalogue.

6 From range to engineering

Propulsion sets speed, lifetime sets range. A rocket that must accelerate *and* brake on one tank with mass ratio R and exhaust speed v_e cruises at $v = c \tanh(v_e \ln R/2c)$ (relativistic rocket equation, half the rapidity on each burn). Its hop range is $r = v\,\tau_{\max}$, where $\tau_{\max}$ is how long the ship, or its crew, survives

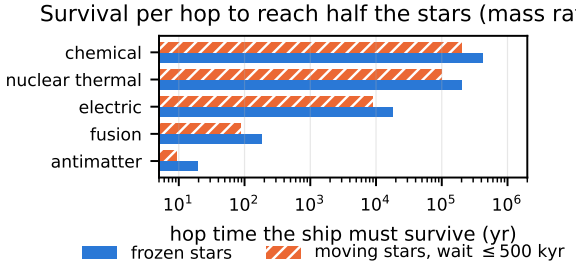


Figure 4: Hop time a ship must survive to reach half the stars within 25 pc, by propulsion technology (mass ratio 10, accelerate and brake on one tank). Hatched bars: colonies may wait up to 500 kyr for stars to drift into range.

in transit. The per-hop survival time needed to percolate is therefore $\tau^* = r_{50}/v$ (Figure 4). At $R = 10$, chemical rockets need 417,195 yr per hop and high-$I_{\rm sp}$ electric propulsion 18,357 yr. A Daedalus-class fusion ship ($v_e = 10^4$ km/s, cruising at 11,507 km/s, 3.8% of c) needs 184 yr, and an idealised antimatter ship 19 yr. If colonies can wait up to 500 kyr for good alignments, every requirement roughly halves (fusion: 88 yr). This turns a technology into a reachability statement: *fusion propulsion and century-scale reliability are enough to percolate through the solar neighbourhood.*

Pilots: greedy fails near threshold, A* is cheap. On the static range graph, a greedy geographic pilot (always hop to the in-range neighbour closest to the target) reaches only 8% of random targets at $r = 2.5$ pc and 49% at 3 pc. It gets stuck in local minima, the "voids" known from wireless routing [4]. It succeeds 99.5% of the time at 4 pc. Near threshold, optimal routes are winding (median stretch 1.78 at 2.5 pc), so a heuristic that never

moves away from the target is wrong by design. A* with the straight-line heuristic finds the same optimal routes as Dijkstra while expanding a median of 10% of Dijkstra's nodes at 2.5 pc and 2% at 4 pc.

Refuelling only where planets are known. Suppose a ship may stop anywhere, with hops of at most 3 pc for crew or hardware reasons, but can refuel only at the 310 known exoplanet hosts within 30 pc. The tank range then needed to reach each star is again a bottleneck, now on a station graph whose edge weights are in-graph shortest distances. We compute it exactly by multi-source Dijkstra and an MST over stations. Reaching half the stars needs a 7.5 pc tank (90%: 10.0 pc), 3.4× the 2.18 pc that suffices when any star can refuel (Figure 5, left). Planet surveys therefore shorten the required tank directly: every newly confirmed host is a potential refuelling station.

Sails need bright targets. A laser-launched sail can leave at 0.2 c, but to stop it must shed its speed on the target's light. Scaling the photogravitational braking limit of [15] (13,800 km/s at α Cen A) as $L^{0.4}$, the median star within 25 pc can stop a sail arriving at only 1,225 km/s. Only 21% can stop one faster than 0.01 c, because most nearby stars are M dwarfs. Sail routes therefore hop between bright stars or crawl into dim ones. With a 1,000-yr hop budget a sail network reaches 39% of stars, against 100% for a rocket at the same cruise speed. At 3,000 yr it reaches 83%, and half the stars take 3,934 yr rather than 336 yr (Figure 5, right).

Exploring everything nearby. Finally we reproduce the classic probe-exploration model of Bjørk [3]. A fleet splits the sky into sectors, and each probe visits every star in its sector, here the 572 stars within 12 pc. Replacing the nearest-neighbour rule with 2-opt local search shortens total flight by 7–9% across fleet sizes of 1–16, consistent with the ∼10% of [7] on synthetic stars (Figure 6). Against the MST lower bound, nearest-neighbour tours are 26% too long and 2-opt tours 15%. At 0.1 c a single probe needs 29.8 kyr to visit them all. A fleet of 8 needs 4.3 kyr. Fleet parallelism, not tour quality, dominates exploration time.

7 Limitations and conclusion

Limitations. (i) *Completeness.* Gaia misses most brown dwarfs and some bright stars. We add the most important ones by hand and bound the remaining effect by injection, which lowers the threshold by about 7%. The true neighbourhood is therefore somewhat *easier* to cross than our graph. (ii) *Kinematics.* Stars move on straight lines. Galactic curvature, binary orbits and close-encounter scattering are ignored. This is adequate for $\lesssim 1$ Myr inside 50 pc but not for Galaxy-scale extrapolation. (iii) *Missing RVs* are drawn from a conditional velocity ellipsoid. Their individual values are uncertain, though the statistics are unbiased. (iv) *Boundaries.* The moving-star results lose inflowing stars after about 500 kyr and are conservative there. (v) *Engineering* models are deliberately simple: one range per ship, refuelling as a binary property of a star, and a sail-braking law scaled from one calibrated case. (vi) We treat a colony's ability to relaunch as given, with an optional settling delay. Sociology is out of scope.

Conclusion. Routed exactly on the measured solar neighbourhood, the interstellar graph has a sharp percolation threshold near 2.16 pc per hop, about twice the typical spacing. That threshold falls by half once ships may wait $\sim 10^5$ years for the stars themselves to close the gaps. In this regime the expansion front moves at stellar speeds. A fusion ship that survives a couple of centuries per hop crosses the neighbourhood. Sails need luminous stopping points, and refuelling only at known planet hosts multiplies the required tank by 3.4. The time-dependent formulation is exact, cheap and scales directly to the 100-pc GCNS and to deeper Gaia releases. Natural next steps are Galactic-potential orbits, gravitational-assist edges between stars, and stochastic colony survival.

Reproducibility. `make test && make all` in `projects/18_interstellar_routes` downloads and caches the catalogue, runs every experiment, and regenerates the figures and every number in this paper (about 3 minutes on a laptop CPU for the 50-pc configuration).

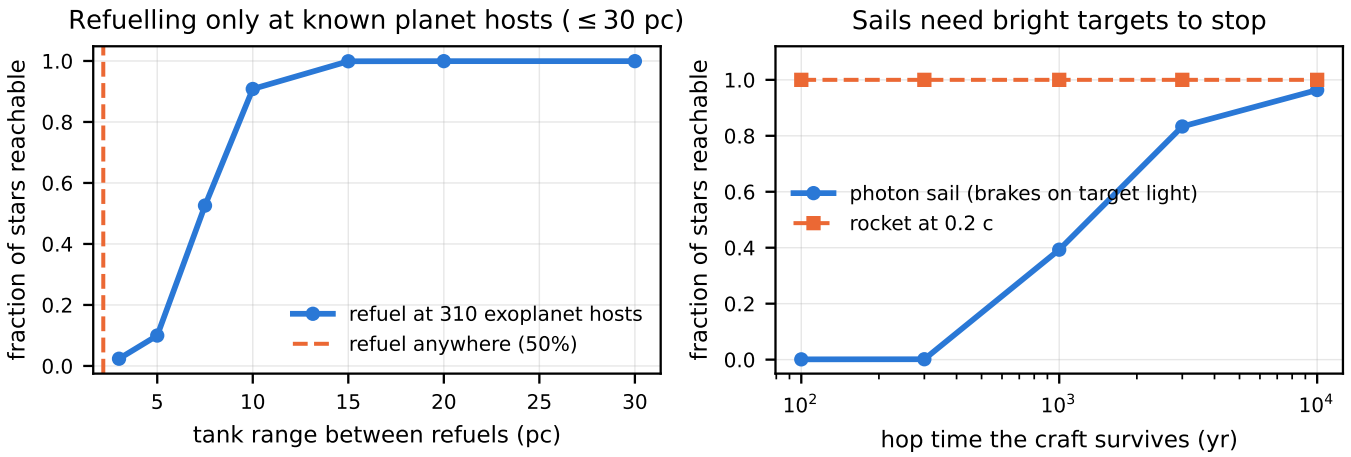


Figure 5: Left: reach within 25 pc against tank range when refuelling is possible only at known exoplanet hosts (hops ≤3 pc). The dashed line is r_{50} when any star can refuel. **Right:** photon sails launched at 0.2 c but braking only on the target star's light, against a rocket flying at the same speed. Reach is plotted against the hop time the craft survives.

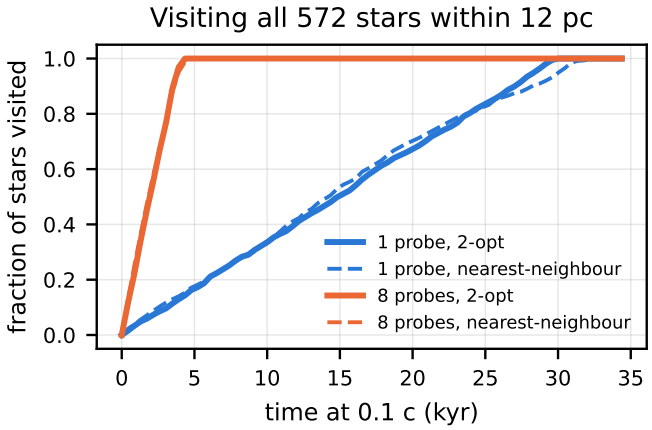


Figure 6: Fraction of stars within 12 pc visited over time by 1 and 8 probes at 0.1 c: 2-opt tours (solid) against the nearest-neighbour rule (dashed).

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